Afnan Hasan

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Education

University of Texas at Austin — Austin, TX
Bachelor of Science in Computer Science — *Aug. 2025 – May 2028*

- Cumulative GPA: **3.96/4.0**
- Relevant Coursework: Data Structures/Algorithms, Computer Architecture, Machine Learning, Operating Systems

Experience

Software Engineering Intern — Oct. 2026 – Present
LotTech — *Austin, TX*

- Built a **real-time voice AI agent** in Python using **Pipecat**, **AssemblyAI** speech-to-text, Rime/Cartesia text-to-speech, and Claude/OpenAI LLMs, answering inbound calls for **35+ dealerships** with response latency minimized to **1,200 ms**
- Implemented **LLM tool calls** for live business-hours checks, service-appointment transportation booking, and follow-up routing to the requested employee, backed by **FastAPI** and **Supabase (PostgreSQL)**
- Hardened a **TypeScript/Hono REST API** to require confirmed-active users before login and requests proceed, and slimmed **Docker images** to install only each service's runtime dependencies

Software Engineering Intern — July 2026 – Sep. 2026
CS Holding — *Part-time, Remote (Milan, Italy)*

- Built a **Next.js dashboard** with email classification triage that handles **200+ emails a day** to centralize daily workflows and task scheduling, using **GitHub Actions** for CI/CD
- Developed a custom **retrieval-augmented generation (RAG) pipeline** using **LangChain** and **pgvector** database for company-wide hybrid search and queries over **3500+ private files** with both semantic and lexical matching

Technical Developer Lead — Jan. 2026 – Present
Texas Convergent — *Austin, TX*

- Co-led the design and development of an all-in-one member portal used by **200+ members** to handle recruiting, attendance, and internal communication
- Built a **fraud-resistant attendance system** using a REST endpoint that generates a short-lived token every 5 seconds, with the display polling for the latest token to render as a QR code, improving **true attendance by 23%**

Software Engineering Intern — June 2025 – Aug. 2025
Lincoln Property Company — *Plano, TX*

- Built an **NLP classifier** to auto-triage maintenance requests by urgency and category, comparing a TF-IDF baseline against a **fine-tuned transformer** and achieving **87% precision** on emergency-priority requests
- Automated weekly expense reporting/visualization for building management using Python, pandas, and matplotlib

Projects

Pantriful | *Java, Spring Boot, PostgreSQL, Flyway, TypeScript, React Native, Docker* — July 2026 – Present

- Built a full-stack kitchen-inventory mobile app with **OAuth2 authentication**, self-issued **JWT session management**, and per-user data isolation, verified by **19+ integration tests** and a full test suite
- Implemented barcode lookup, low-stock notifications, and receipt-scanning to reduce manual pantry updates
- Built a custom **recipe recommendation engine** matching a user's live pantry against cooking history, using **schema-constrained LLM output** to suggest healthy recipes for target users (college students)

Homegrown | *TypeScript, React Native, PostgreSQL, TensorFlow* — Sep. 2025 – May 2026

- Built a full-stack marketplace app for produce, plant posts, and community groups, growing to **30+ early users**
- Built price-suggestion model for produce using **polynomial regression**, trained on **15,000+ historical listings** and current market trends for maximized profit for sellers using the app
- Trained a **MobileNetV2** classifier using transfer learning on **40,000+ images** for produce freshness verification, reaching **94% test accuracy**

MyDrive | *Java, Spring Boot, React, TypeScript, Docker, AWS* — May 2025 – Aug. 2025

- Built a full-stack car rental platform with a Spring Boot/Java REST API, implementing JWT-based authentication, fleet/reservation management, and **Stripe payment processing** used by ~**85 monthly renters** across **45+ vehicles**
- Implemented booking workflows requiring license and insurance verification and lease signing before payment, and designed reservation logic to prevent scheduling conflicts across the fleet

Technical Skills

Languages: Java, Python, C, C++, JavaScript, TypeScript, SQL, HTML, CSS
Frameworks: Spring Boot, React, React Native, Next.js, FastAPI, Hono
Databases & Cloud: PostgreSQL, Supabase, pgvector, AWS, Docker, Kubernetes, GitHub Actions, Git
Libraries: pandas, NumPy, TensorFlow, PyTorch, Matplotlib, Streamlit, scikit-learn
AI/ML: LangChain, LangGraph, RAG, NLP, Pipecat, AssemblyAI, Claude API, OpenAI API